# Amazon Stock Analysis Report

The analysis of the Amazon (AMZN) stock dataset (1997-2021) reveals a remarkable growth story characterized by extreme volatility and long-term resilience.

## 1. Key Performance Metrics

The stock has shown extraordinary long-term returns since its IPO in 1997.

- **Total Return:** 173,322.49% (from May 1997 to 2021).
- **Compound Annual Growth Rate (CAGR):** 35.71%. This level of sustained annual growth over 24 years is extremely rare in the stock market.
- **Recent Performance (Last 5 Years):** The stock maintained strong momentum with a **315% total return** and a CAGR of **32.93%**.
- **Annualized Volatility:** 57.89%. This indicates that while the returns are high, the stock experiences significant price swings.
- **Maximum Drawdown:** -94.40%. This occurred during the dot-com bubble crash (late 1999 to 2001), showing the stock's survival through extreme market corrections.

## 2. Price Trend & Moving Averages

The long-term chart (see amazon_price_trend.png) shows a parabolic rise, particularly after 2015.

- **Support & Momentum:** The 50-day and 200-day moving averages (MAs) act as significant trend indicators. In the last 5 years (amazon_recent_trend.png), the stock price consistently stayed above the 200-day MA, indicating a strong long-term bullish trend.

- **Recent Consolidation:** Toward the end of the dataset (2020-2021), the stock saw higher volatility but continued to push toward all-time highs above $3,700.

## 3. Seasonality and Returns Distribution

- **Strongest Months:** Historically, **April (0.47%)** and **November (0.35%)** have the highest average daily returns.

- **Weakest Month: February (-0.13%)** is the only month with a negative average daily return.

- **Daily Returns:** Most daily returns are concentrated around 0%, but the distribution has "fat tails," meaning extreme price jumps or drops are more common than in a normal distribution (see amazon_returns_dist.png).

    - **Best Day:** 34.47% (Nov 26, 2001).

    - **Worst Day:** -24.77% (July 24, 2001).

**4. Key Insights**

1. **High-Risk, High-Reward:** Amazon has been one of the best-performing assets in history, but it required investors to endure a nearly 95% loss during its early years to reap the ultimate rewards.

2. **Growth Acceleration:** The slope of the price curve steepened significantly after 2015, marking its transition from a dominant e-retailer to a cloud computing (AWS) and services powerhouse.

3. **Volume Trends:** Trading volume was exceptionally high in the early 2000s relative to price. In recent years, while the nominal volume is lower, the dollar value of trades has exploded due to the high share price.

**5. Historical Stagnation Period (2000–2008)**

While the overall trend is exponential, the notebook highlights a specific **9-year "Lost Decade"** for Amazon:

- **Insight:** From the start of 2000 to the end of 2008, the stock actually saw a cumulative decline of **-42.62%**.

- **Context:** This period includes the dot-com bust and the 2008 financial crisis. The notebook identifies this as a "stagnation phase" before the rapid acceleration began in 2009.

**6. Rolling Correlation Analysis**

The notebook provides a more dynamic view of the relationship between **Volume** and **Price**:

- **Finding:** Instead of a single correlation value, the 30-day rolling correlation shows that the link between volume and price is highly unstable.

- **Insight:** There are periods where price and volume are perfectly correlated (+1.0) and others where they are perfectly inversely correlated (-1.0). This suggests that volume spikes in Amazon's history aren't always "buying" spikes; they often coincide with panic selling or neutral consolidation phases.

**7. Intraday Price Delta (Close - Open)**

The notebook analyzes the absolute intraday price change rather than just daily percentage returns:

- **Insight:** By comparing Close - Open against Volume, the notebook observes how much "conviction" exists within a single trading day. High-volume days with small price deltas indicate intense "battles" between buyers and sellers (high liquidity), whereas high-volume days with large deltas indicate strong directional breakouts.

**8. Annual Performance Breakdown**

The notebook visualizes annual returns in a bar chart format (see annual_returns_bar.png), highlighting extreme outliers:

- **1998 Peak:** A massive **979.8%** return in a single year during the early expansion.
- **2000 Bottom:** A brutal **-82.5%** return during the dot-com crash.
- **Consistent Growth:** Since 2015, every year (up to 2021) has remained positive, with 2020 showing a significant **71.6%** surge due to the e-commerce boom during the pandemic.

**9. Feature Correlation Matrix**

The notebook includes a comprehensive correlation heatmap for all price features:

- **Key Finding:** There is a near-perfect correlation (>0.99) between **Open, High, Low, and Close** prices, which is expected for a high-liquidity stock.
- **Volume Decoupling:** Volume has a very low correlation with the absolute price level, confirming that Amazon's price appreciation was driven by valuation growth rather than just an increase in trading activity.

**10. Market Correction Identifiers**

The notebook specifically notes a "market correction" during the **2017–2018** period. While the year ended positive overall, the notebook's visual analysis points out the significant volatility and mid-year drawdowns that occurred as the stock's growth trajectory shifted into a steeper, exponential phase.

Summary Table

| Metric | Value |
|---|---|

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| --- | --- |
| **Start Price** | $1.40 (1997) |
| **End Price (Max)** | $3,731.41 (2021) |
| **CAGR (Total)** | 35.71% |
| **Volatility (Annualized)** | 57.89% |
| **Best Month** | April |
| **Worst Month** | February |